

Computer Networks - SET 1

◆ Section 1: Introduction & Fundamentals

1. Which topology connects all nodes to a single central device?

- a) Ring
- b) Mesh
- c) Star
- d) Bus

Answer: c) Star

Explanation: In a star topology, all devices are connected to a central hub or switch.

2. Which device operates only at the physical layer?

- a) Router
- b) Switch
- c) Repeater
- d) Bridge

Answer: c) Repeater

Explanation: Repeaters operate at Layer 1 to boost signal strength over long distances.

3. What is bandwidth in networking?

- a) Time delay in a network
- b) Amount of data that can be transmitted in a given time
- c) Distance between two routers
- d) Signal frequency range

Answer: b) Amount of data that can be transmitted in a given time

Explanation: Bandwidth is the data transfer capacity, usually measured in Mbps or Gbps.

4. Which of the following defines network latency?

- a) Maximum bandwidth of a connection
- b) Time delay in data transmission
- c) Number of users on a network
- d) Amount of data loss during transfer

Answer: b) Time delay in data transmission

Explanation:

It is going to be hard but, hard does not mean impossible.

Latency is the **time it takes for data to travel** from source to destination. It's measured in milliseconds and affects real-time applications like video calls and gaming.

◆ **Section 2: OSI & TCP/IP Models**

5. Which layer of the OSI model ensures reliable delivery of data?

- a) Data Link
- b) Network
- c) Transport
- d) Application

Answer: c) Transport

Explanation: The transport layer ensures error checking, flow control, and reliable delivery.

6. In the OSI model, which layer converts data to signals?

- a) Network
- b) Data Link
- c) Physical
- d) Session

Answer: c) Physical

Explanation: The physical layer converts bits into electrical or optical signals.

7. Which layer handles IP addressing and routing?

- a) Data Link
- b) Network
- c) Session
- d) Transport

Answer: b) Network

Explanation: The network layer manages logical addressing and routing between networks.

8. What is the function of the Session layer in OSI?

- a) Manages encryption
- b) Establishes, maintains, and terminates connections
- c) Routes packets
- d) Converts data formats

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Answer: b) Establishes, maintains, and terminates connections

Explanation: The session layer is responsible for managing and controlling dialog sessions.

◆ **Section 3: IP Addressing & Protocols**

9. What is the default subnet mask for a Class C IP address?

- a) 255.0.0.0
- b) 255.255.0.0
- c) 255.255.255.0
- d) 255.255.255.255

Answer: c) 255.255.255.0

Explanation: Class C addresses use the first 3 octets for network ID.

10. Which protocol assigns IP addresses dynamically?

- a) DNS
- b) FTP
- c) DHCP
- d) HTTP

Answer: c) DHCP

Explanation: DHCP provides IP addresses and settings to devices automatically.

11. What does ARP do in a network?

- a) Maps MAC to IP
- b) Maps IP to MAC
- c) Authenticates users
- d) Provides DNS resolution

Answer: b) Maps IP to MAC

Explanation: ARP finds the MAC address for a given IP.

12. What is CIDR used for in IP addressing?

- a) Assigning MAC addresses
- b) Classifying protocols
- c) More efficient IP address allocation
- d) Routing based on hostname

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Answer: c) More efficient IP address allocation

Explanation: CIDR allows flexible subnetting beyond class-based IPs.

13. What is the primary advantage of IPv6 over IPv4?

- a) Lower latency
- b) More secure
- c) Larger address space
- d) Easier to configure

Answer: c) Larger address space

Explanation: IPv6 offers 128-bit addresses for virtually unlimited devices.

◆ Section 4: Switching & Routing

14. What kind of device breaks a broadcast domain?

- a) Hub
- b) Repeater
- c) Switch
- d) Router

Answer: d) Router

Explanation: Routers separate broadcast domains by default.

15. Which of these uses store-and-forward switching?

- a) Hub
- b) Switch
- c) Bridge
- d) Repeater

Answer: b) Switch

Explanation: Switches read the entire frame before forwarding.

16. Which routing protocol uses hop count as a metric?

- a) OSPF
- b) BGP
- c) RIP
- d) EIGRP

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Answer: c) RIP

Explanation: RIP uses hop count to determine the best route.

17. Which protocol helps detect network path failures and reroutes traffic?

- a) DHCP
- b) ICMP
- c) OSPF
- d) SNMP

Answer: c) OSPF

Explanation: OSPF is a link-state routing protocol that adjusts to path changes.

18. What is the primary function of NAT?

- a) Encrypt data
- b) Allow private IPs to communicate with the internet
- c) Assign MAC addresses
- d) Increase bandwidth

Answer: b) Allow private IPs to communicate with the internet

Explanation: NAT translates internal private IPs to public IPs for internet access.

◆ **Section 5: TCP/UDP & Ports**

19. Which protocol guarantees reliable delivery?

- a) UDP
- b) TCP
- c) ICMP
- d) ARP

Answer: b) TCP

Explanation: TCP uses acknowledgments and checks for reliability.

20. Which protocol is connectionless and used for streaming?

- a) FTP
- b) TCP
- c) UDP
- d) SSH

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Answer: c) UDP

Explanation: UDP is lightweight and does not require connection setup.

21. What is the default port number for HTTPS?

- a) 21
- b) 25
- c) 443
- d) 80

Answer: c) 443

Explanation: Port 443 is used for secure HTTP traffic.

22. Which port number is used by SSH?

- a) 20
- b) 22
- c) 80
- d) 53

Answer: b) 22

Explanation: SSH uses port 22 for encrypted remote access.

◆ **Section 6: Network Devices**

23. Which device connects two different networks?

- a) Switch
- b) Hub
- c) Router
- d) Repeater

Answer: c) Router

Explanation: Routers are used for inter-network communication.

24. What does a switch use to forward data?

- a) IP address
- b) Port number
- c) MAC address
- d) Protocol ID

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Answer: c) MAC address

Explanation: Switches work at Layer 2 and use MAC addresses.

25. What device is used to convert digital signals to analog?

- a) Router
- b) Bridge
- c) Modem
- d) Repeater

Answer: c) Modem

Explanation: Modems modulate/demodulate digital signals over analog lines.

26. What device operates at both Layer 2 and Layer 3?

- a) Bridge
- b) Router
- c) Layer 3 Switch
- d) Hub

Answer: c) Layer 3 Switch

Explanation: Layer 3 switches can route like routers and switch like switches.

◆ **Section 7: Wireless & LAN Technologies**

27. Which protocol is used for secure remote login?

- a) Telnet
- b) SSH
- c) FTP
- d) SNMP

Answer: b) SSH

Explanation: SSH is a secure alternative to Telnet.

28. What does SSID stand for in wireless networks?

- a) Secure Socket Identifier
- b) Service Set Identifier
- c) Server Site ID
- d) Signal Stream Identifier

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Answer: b) Service Set Identifier

Explanation: SSID is the name of a Wi-Fi network.

29. Which standard defines Wi-Fi?

- a) 802.3
- b) 802.11
- c) 802.5
- d) 802.15

Answer: b) 802.11

Explanation: 802.11 is the IEEE standard for wireless LAN.

30. What does a VPN provide?

- a) More bandwidth
- b) Static routing
- c) Secure communication over the internet
- d) Dynamic IP allocation

Answer: c) Secure communication over the internet

Explanation: VPNs encrypt and tunnel traffic securely.

◆ **Section 8: Application Layer Protocols**

31. Which protocol is used to transfer web pages?

- a) FTP
- b) DNS
- c) HTTP
- d) DHCP

Answer: c) HTTP

Explanation: HTTP is used to access web content.

32. What is the purpose of DNS?

- a) Encrypt data
- b) Assign IP addresses
- c) Map domain names to IP addresses
- d) Check for packet loss

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Answer: c) Map domain names to IP addresses

Explanation: DNS translates human-friendly names to IP addresses.

33. What protocol is used to send emails?

- a) FTP
- b) SMTP
- c) POP3
- d) DHCP

Answer: b) SMTP

Explanation: SMTP sends outgoing emails.

34. What port is typically used by DNS?

- a) 80
- b) 25
- c) 53
- d) 110

Answer: c) 53

Explanation: DNS commonly uses port 53 (UDP/TCP).

◆ **Section 9: Network Security Basics**

35. Which of the following is not a function of a firewall?

- a) Packet filtering
- b) Traffic monitoring
- c) IP addressing
- d) Access control

Answer: c) IP addressing

Explanation: IP addressing is not related to firewall functions.

36. Which attack overwhelms a system with traffic?

- a) Phishing
- b) Spoofing
- c) DDoS
- d) Sniffing

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Answer: c) DDoS

Explanation: DDoS floods the network with requests.

37. Which tool is used to analyze network traffic?

- a) Netstat
- b) Wireshark
- c) Ping
- d) FTP

Answer: b) Wireshark

Explanation: Wireshark captures and inspects packet data.

38. Which of the following is a QoS feature?

- a) ARP resolution
- b) Traffic prioritization
- c) NAT translation
- d) Port mirroring

Answer: b) Traffic prioritization

Explanation: QoS prioritizes critical data like VoIP.

◆ **Section 10: Real-World Commands & Utilities**

39. What does the ping command test?

- a) Port connectivity
- b) DNS resolution
- c) Network reachability
- d) Bandwidth

Answer: c) Network reachability

Explanation: ping checks if a host is reachable.

40. Which command will assign a static IP to a Linux machine temporarily?

- a) ip set address
- b) netstat set ip
- c) ip addr add
- d) route static set

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Answer: c) ip addr add

Explanation: The ip addr add command is used to assign IPs temporarily.

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